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ORIGINAL PAPER

# Effects of RANS Turbulence Models on Aerodynamics of Slender-Bodied Launch Vehicles with Protuberance

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## Abstract

A slender-bodied vehicle with asymmetrically arranged protuberance generates strong side force due to asymmetric vortices, even at a low angle-of-attack. We investigated effects of the well-known RANS turbulence models [SA-R ($C_{\mathrm{rot}} = 0.0$, 1.0, and 2.0), SST, and SST-2003] by comparing the numerically obtained side force values on supersonic slender-body, along with the flow structure. As a result, all the SA-R models showed good agreement with the experiment regardless of the $C_{\mathrm{rot}}$ (which controls the degree of modification from the original SA model), although a separation point on the protuberance side slightly changed depending on the $C_{\mathrm{rot}}$ value. On the other hand, as for the SST models, when the vorticity was used to evaluate eddy viscosity (original SST) the side force exhibit 44% deviation from the experiment, whereas SST-2003, in which the strain rate was employed instead, significantly reduced the discrepancy to 0.7%.

**Keywords** Turbulent flow · Vortex · Numerical analysis · Supersonic flow

## 1 Introduction

In recent years, the development of small and nanosatellites has been actively pursued [1, 2]. In the future, such a demand is expected to increase further with the development of information technology including the Internet of Things (IoT) [3]. As a result, the need for low-cost and high-frequency launches is increasing, and the space transport vehicles are being downsized [1, 2]. As for the rocket, various devices are attached to the surface of the vehicle to secure the internal volume as the size of the body is reduced. These protuberances form asymmetric vortex structure in the leeward side of the vehicle. Therefore, the side force, which acts laterally to the flight direction of the vehicle, is generated even at a relatively low angle of attack ($\alpha < 15°$) [4]. The side force can prevent a launch vehicle from flying its planned path. Because of this, it is very important to understand the effect of protuberances on the aerodynamic characteristics, especially the side force, to achieve safe and efficient launch.

The side force acting on the slender-bodied vehicle with protuberances and the ambient flow are characterized by the behaviour of wake vortices of protuberances or flow separation from the body [4]. Since the vortex core generally possesses low pressure, if the wake vortex of the protuberance is generated near the body, the vehicle is pulled strongly by the vortex [5]. Moreover, in the case where the protuberance is attached to only one side of the body, the wake vortex develops toward the aft portion and then moves further away from the body. In such a case, a large side force acts from the side with the protuberance to the side without it because of the pressure difference between both sides of the vehicle.

Thus, the aerodynamic characteristics of the launch vehicle varies greatly due to the behaviour of the vortex, depending on the protuberance positions. Consequently, it is essential to investigate aerodynamic characteristics and ambient flow field by utilizing wind tunnel test and numerical analysis to realize the safe launch of the vehicle. As a concrete example, the prediction of flow field and aerodynamic forces acting on the flying vehicle with protuberances utilizing computational fluid dynamics (CFD) has been widely performed [4–6]. Most of them are calculated by solving

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(a) Side view. (b) Front view.

**Fig. 1** Model for calculations

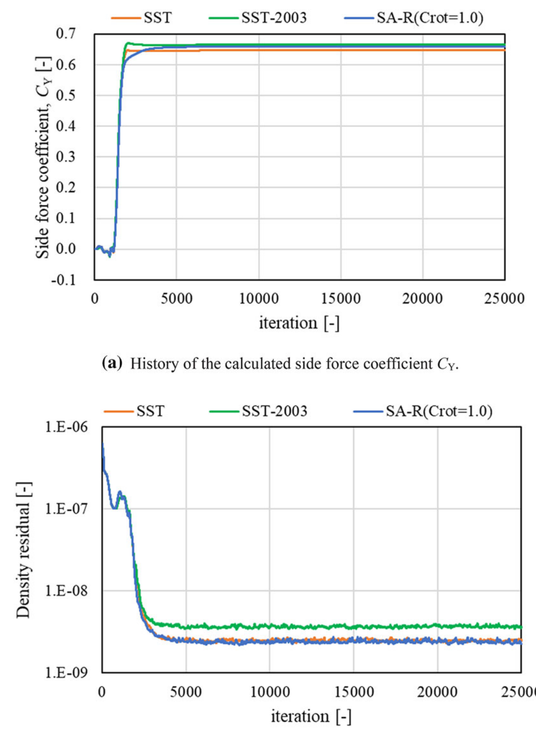
RANS (Reynolds Averaged Navier–Stokes equations), and the results may depend on the selected turbulence models. In fact, a study [7] on delta-wing, in which separation vortices are generated on the leeward of the object as in the case of the slender body, reported that the prediction accuracy of flow separation from leading edge and vortex breakdown at a high angle of attack significantly varied depending on the turbulence models.

Therefore, in this study, the effects of turbulence models on numerical simulations of launch vehicle with protuberances are investigated. Numerical calculations are performed using the SA-R models [8, 9] (SA: Spalart–Allmaras), which are one-equation models widely used in the aerospace field, and the SST models (SST: Shear Stress Transport) [10, 11], which are two-equation models. Furthermore, for the SA-R, it is reported in the literature [12] that different results are obtained depending on the value of the modification parameter $C_{\mathrm{rot}}$ (as detailed in Sect. 2.3.1) from the original SA [13] (to accommodate flow with curvature). However, the $C_{\mathrm{rot}}$ dependence on the calculated aerodynamic forces acting on the vehicle, including side force, has not been investigated enough. Hence, we also investigate the $\mathrm{C_{rot}}$ dependence on the side force and the ambient flow physics around the launch vehicle in supersonic flow.

# 2 Numerical Setup

## 2.1 Model for Calculation

The model for this work is shown in Fig. 1. This model has the same geometry as that used by Kawauchi et al. [4] in their investigation of the effect of protuberances on the surface of the launch vehicle on induced side forces, and consists of a slender body and a protuberance attached over its surface. The protuberance mimics the size of the attitude control device of the Epsilon launch vehicle [6]. The slender body has a total length of $L = 368$ mm and a diameter of $D = 41.5$ mm. The length of the protuberance is 0.06L, and the protrusion from the body is 0.15D. This protuberance is attached to the leeward side (45° to port side from the vertical plane of the body) at the front portion of the vehicle (body axial position: $x = 0.22$ L). In the case where the protuberance is attached to this position, a wake vortex of the protuberance develops toward the aft of the vehicle, and as a result, a strong side force is generated from the protuberance side to the smooth side [4].




(a) History of the calculated side force coefficient $C_Y$.

(b) History of the density residual.

**Fig. 2** Iteration history of SST, SST-2003 and SA-R ($C_{\mathrm{rot}} = 1.0$)

## 2.2 Calculation Methods

Numerical calculations are performed using FaSTAR [14], a fluid analysis solver developed by JAXA. The governing equations are three-dimensional compressible RANS, which are discretized by the cell-centered finite volume method and solved with spatial quadratic accuracy. A numerical flux for an inviscid term is calculated by SLAU [15]. The gradient is calculated by the Green-Gauss method [16]. The gradient limiting function is minmod [17]. The LU-SGS method [18] is used as the time integration method, and steady-state calculations were performed with the Courant number of 50. Figure 2 shows the history of the side force coefficient $C_{\mathrm{Y}}$ and density residuals for calculations by each model. (For the SA-R model, the history for $C_{\mathrm{rot}} = 1.0$ is included as a representative example.) According to the results, the values of the side force coefficient $C_{\mathrm{Y}}$ are well converged, and the density residual falls by more than two orders of magnitude from the initial iteration of calculations. From these reasons, the convergence of the solutions obtained by computations in this study is verified.

## 2.3 Turbulent Models

### 2.3.1 SA-noft2 and SA-noft2-R

The SA and SA-R models are one-equation models that solve the following equation. In this work, we use SA-noft2 [19], which omits the $f_{t2}$ term (= tripping term for controlled transition) from the original model (but simply denoted as "SA" for brevity in the rest of this work).

$$\frac{\partial\widetilde{\upsilon}}{\partial t}+\frac{\partial\bar{u}_j\widetilde{\upsilon}}{\partial x_j}=C_{b1}\left[1-f_{t2}\right]\widetilde{S}\widetilde{\upsilon}+\frac{1}{Re}\frac{1}{\sigma}\frac{\partial}{\partial x_j}\left[(\upsilon+\widetilde{\upsilon})\frac{\partial\widetilde{\upsilon}}{\partial x_j}\right]-\frac{1}{Re}C_{W1}f_W\left[\frac{\widetilde{\upsilon}}{d}\right]^2+\frac{1}{Re}\frac{C_{b2}}{\sigma}\frac{\partial\widetilde{\upsilon}}{\partial x_j}\frac{\partial\widetilde{\upsilon}}{\partial x_j}. \quad (1)$$

$\tilde{\upsilon}$ is a working variable and is used to obtain the turbulent viscosity $\mu_t$.

The Reynolds stress $\overline{\tau}^t_{ij}$ and turbulent viscosity $\mu_t$ are defined as follows:

$$\overline{\tau}^t_{ij}=-\bar{u}'_i u'_j=\frac{1}{Re}\mu_t\left[\left(\frac{\partial\bar{u}_i}{\partial x_j}+\frac{\partial\bar{u}_j}{\partial x_i}\right)-\frac{2}{3}\frac{\partial\bar{u}_k}{\partial x_k}\delta_{ij}\right], \quad (2)$$

$$\mu_t=\bar{\rho}\tilde{\upsilon}f_{V1}. \quad (3)$$

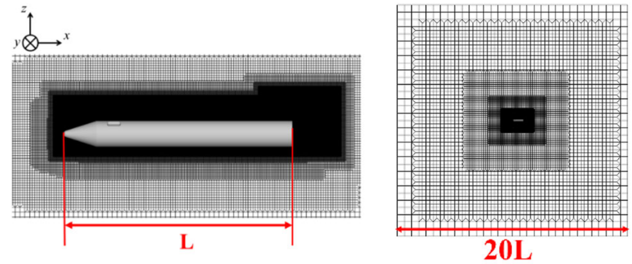
See Ref. [8, 9] for Auxiliary functions and constants.

The original SA model has the disadvantage that it is difficult to distinguish "vortices due to flow around a curved wall" from "vortices due to turbulence flow". To remedy this problem, in the SA-R model, the magnitude of vorticity $|\Omega|$ is replaced with the following:

$$|\Omega|\rightarrow|\Omega|+C_{rot}\min(0,|S|-|\Omega|), \quad (4)$$

$$|S|=\sqrt{2S_{ij}S_{ij}},\ S_{ij}=\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right), \quad (5)$$

where $|S|$ is the strain rate. With the above improvements, the SA-R model suppresses turbulent viscosity by allowing the vorticity to exceed the strain rate at region where the effect of rotation is more dominant than that of turbulence. The parameter $C_{rot}$, which controls the degree of modification from original SA to SA-R as shown in Eq. (4), has been arbitrarily determined by users. NASA's turbulence modelling resource [20] recommends $C_{rot}=2.0$, while Lei [7] states that $C_{rot}=2.0$ is physically incorrect and $C_{rot}=1.0$ is most appropriate. As such, further discussion is needed regarding the adoption of the $C_{rot}$ value. Therefore, in this work, numerical calculations are performed for three different values of $C_{rot}$: $C_{rot}=0.0$ (original SA [13]), 1.0 [7], 2.0 [20].




**(a)** Body vicinity. **(b)** Overall view.

**Fig. 3** Computational Grid

### 2.3.2 SST and SST-2003

The Menter-SST is a two-equation model that solves the Wilcox $k$-$\omega$ equation in the vicinity of the wall and uses the standard $k$-$\varepsilon$ model in the fully turbulent region away from the wall. In this model, the following equations are solved.

$$\frac{\partial\bar{\rho}k}{\partial t}+\frac{\partial\bar{\rho}\bar{u}_j k}{\partial x_j}=P_k-\beta^*\bar{\rho}\omega k+\frac{1}{Re}\frac{\partial}{\partial x_j}\left[(\mu+\sigma_k\mu_t)\frac{\partial k}{\partial x_j}\right],$$

$$\frac{\partial\bar{\rho}\omega}{\partial t}+\frac{\partial\bar{\rho}\bar{u}_j\omega}{\partial x_j}=\gamma\frac{1}{\upsilon_t}P_k-\beta^*\bar{\rho}\omega^2+\frac{1}{Re}\frac{\partial}{\partial x_j}\left[(\mu+\sigma_k\mu_t)\frac{\partial k}{\partial x_j}\right]+2(1-F_1)\bar{\rho}\sigma_{\omega2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial\omega}{\partial x_j}, \quad (6)$$

where $k$ is the turbulent kinetic energy and $\omega$ is the energy dissipation rate.

The Reynolds stress $\overline{\tau}^t_{ij}$ and turbulent viscosity $\mu_t$ are defined as follows:

$$\overline{\tau}^t_{ij}=-\overline{u'_i u'_j}=\frac{1}{Re}\mu_t\left[\left(\frac{\partial\bar{u}_i}{\partial x_j}+\frac{\partial\bar{u}_j}{\partial x_i}\right)-\frac{2}{3}\frac{\partial\bar{u}_k}{\partial x_k}\delta_{ij}\right]-\frac{2}{3}\bar{\rho}k\delta_{ij}, \quad (7)$$

$$\mu_t=\min\left(\frac{k}{\omega},\frac{a_1 k}{\Omega F_2}\right). \quad (8)$$

See Ref. [10] for details on blending functions $F_1$ and $F_2$, for connecting the $k$-$\varepsilon$ and $k$-$\omega$ models and auxiliary constants. In the original SST model [10], vorticity $\Omega$ is used to evaluate turbulent viscosity $\mu_t$ as shown in Eq. (8). However, F.R. Menter, the developer of SST, made an improvement using the strain rate $S$ instead of vorticity $\Omega$, as shown below [11]. A numerical analysis using this improved version of the model, SST-2003, is also performed in this work.

$$\mu_t=min\left(\frac{k}{\omega},\frac{a_1 k}{SF_2}\right). \quad (9)$$

As for the production term $P_k$ in the transport equations, the limiter suggested by Menter is applied in both SST and SST-2003 models as follows:

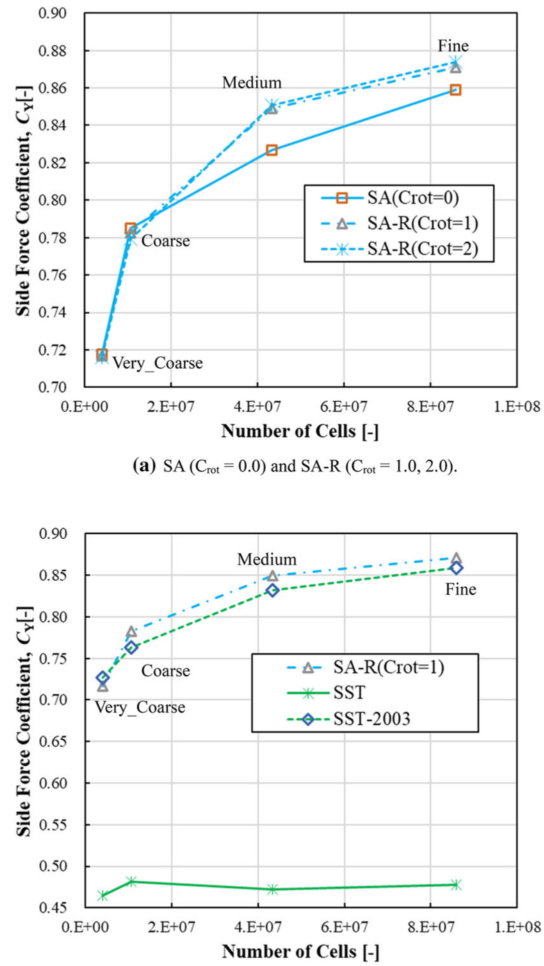
$$\mathrm{P_k} = min\left(\frac{1}{\mathrm{Re}}\mu_t \mathrm{S}^2, 20\rho k\omega\right) = \min\left(\frac{1}{\mathrm{Re}}\mu_t \mathrm{S}^2, 20\rho\varepsilon\right). \tag{10}$$

### 2.4 Computational Grid

The computational grid in this work is shown in Fig. 3. HexaGrid [21], a hexahedron-based automatic grid generation tool, is used to generate the grid. The computational domain is a 20L × 20L × 20L cube, and composed of (mostly) hexahedrons, along with prisms, pyramids, and tetrahedrons. The number of cells is about 44 million, and the-grid spacing closest to the wall is $1.0 \times 10^{-3}$ mm ($y^+{}_{max} = 0.86$). The number of cell layers adjacent to the wall for resolving the viscous boundary layer is approximately 20, and the number of face cells over the wall is approximately 265,000. To investigate the grid dependence of each turbulence model, four different grids (Very_Coarse, 4 million cells; Coarse, 10 million cells; Medium, 43 million cells; Fine, 86 million cells) were prepared and numerical calculations were performed under the conditions of $\alpha = 15°$ and Mach 1.5. The obtained results for the side force coefficient $C_Y$ are shown in Fig. 4. In Fig. 4a, the effect of $C_{rot}$ on the side force coefficient $C_Y$ is small for the two coarser grids (Very_Coarse and Coarse); difference in $C_Y$ depending upon $C_{rot}$ value is at most 0.38%. Meanwhile, the calculated results using finer grids (Medium and Fine) show increasing differences for each $C_{rot}$ value; the largest difference in the side force coefficient $C_Y$ with $C_{rot}$ value is 2.7%. Hence, to capture the effect of the $C_{rot}$ value, it is appropriate to use a Medium or Fine grid. As for the SST and SST-2003 results (Fig. 4b), the SST results are not significantly affected by the grid density (the largest difference in $C_Y$ is 3.4% between Very_Coarse and Coarse). Furthermore, with respect to the difference of $C_Y$ obtained from the numerical calculation using SST-2003, the differences in calculated $C_Y$ of SST-2003 are 4.9% between Very_Coarse and Coarse, 9.1% between Coarse and Medium, and 3.2% between Medium and Fine. Therefore, the side force coefficient $C_Y$ for the Medium grid is considered to be converged. For these reasons, the Medium grid was chosen for this study.

### 2.5 Computational Condition

The uniform flow Mach number is $M_\infty = 1.5$ and the Reynolds number based on the body length $L$ is $1.1 \times 10^7$. In addition, the previous study [4] showed that when the angle of attack $\alpha$ is less than 15°, the side force is nearly proportional to the angle of attack $\alpha$, and that when the angle of attack is between 15° and 20°, the side force is almost constant with respect to $\alpha$. Based on this tendency, the angle of attack $\alpha = 15°$ is selected here.




(a) SA ($C_{rot}$ = 0.0) and SA-R ($C_{rot}$ = 1.0, 2.0).

(b) SST (-2003) and SA-R ($C_{rot}$ = 1.0).

**Fig. 4** Verification of grid dependence

## 3 Results and Discussions

### 3.1 Comparison of Flow Field

First, as an overview of the flow structure around the vehicle, Fig. 5 shows the visualization results of the Q iso-surfaces calculated by the SA-R ($C_{rot} = 1.0$) model. The wake vortices of the protuberance develop toward the aft portion of the vehicle, and the separation vortices on the leeward side of the vehicle on the starboard side (hereafter referred to as the smooth side), where there are no protuberances, can be seen. We show in Fig. 6 the vorticity (left) and the turbulent viscosity (right) distributions around the vehicle, to compare the overall flow fields. For the SA ($C_{rot} = 0.0$) and SA-R ($C_{rot}$

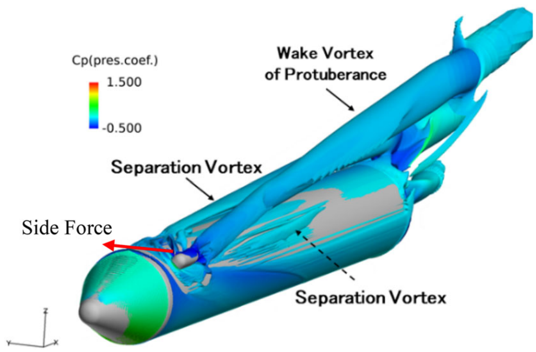
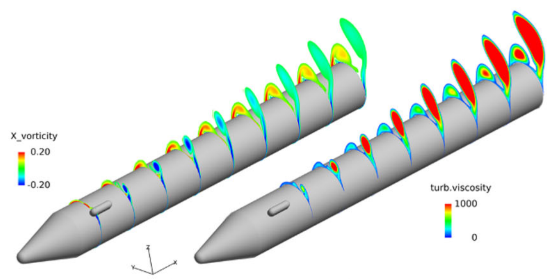



**Fig. 5** Overview of the flow field (SA-R, $C_{\mathrm{rot}} = 1.0$)

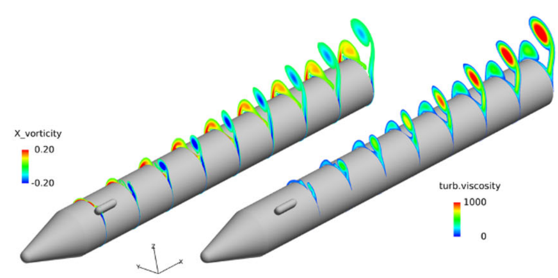
$= 1.0, 2.0$), as the value of $C_{\mathrm{rot}}$ increases, the vorticity of the protuberance wake vortex also develops. As for the SST models, in the case of the calculation results of the original SST, the vorticity on both sides of the vehicle is weak and strong turbulent viscosity is generated throughout the leeward side. In the case of the SST-2003, the vorticity on both sides is stronger and the turbulent viscosity is suppressed compared to the original SST.

## 3.2 Comparison of Side Force Coefficient

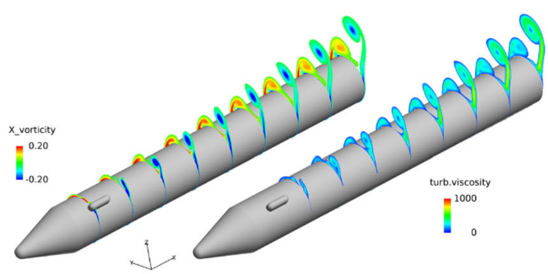
Figure 7 shows the calculated and experimental results of the side force coefficients $C_{\mathrm{Y}}$ for each turbulence model. (Note that the positive side force acts from the protuberance side to the smooth side.) The corresponding wind tunnel tests [4] were performed at Sagamihara Campus' supersonic wind tunnel in Institute of Space and Astronautical Science (ISAS), Japan Aerospace Exploration Agency (JAXA). This wind tunnel is a blowdown-to-atmosphere facility using air from a pressurized plenum as the test gas flowing through a 600 mm × 600 mm square test section, and force measurements were made using an interior balance (manufactured by Nissho-Electric-Works Co., Ltd.). According to the results,



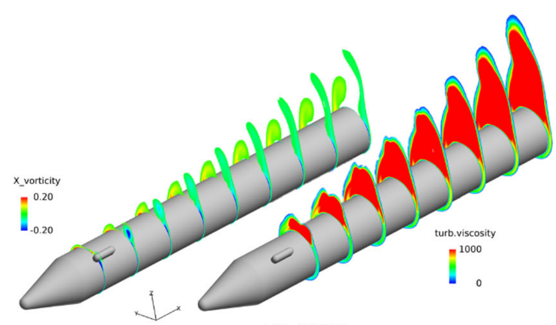

**(a)** $C_{\mathrm{rot}} = 0.0$ (SA)



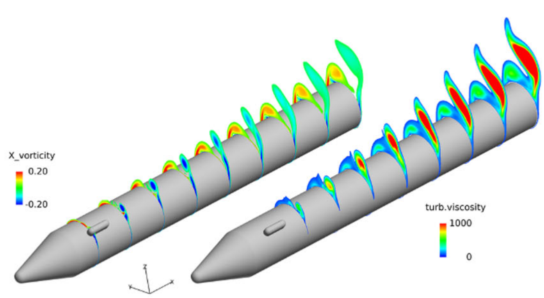

**(b)** $C_{\mathrm{rot}} = 1.0$ (SA-R)




**(c)** $C_{\mathrm{rot}} = 2.0$ (SA-R)




**(d)** SST




**(e)** SST-2003

**Fig. 6** Visualization results of vorticity (left) and turbulent viscosity (right)

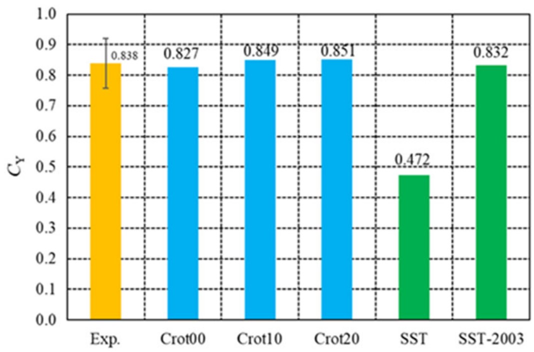



**Fig. 7** Side force coefficients $C_Y$

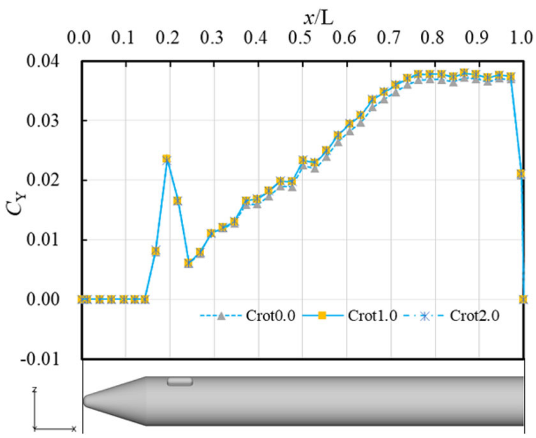



**Fig. 8** Local side force distribution in the axial direction for SA ($C_{rot}$ = 0.0) and SA-R ($C_{rot}$ = 1.0, 2.0)

the experimental and calculated results for the SA ($C_{rot}$ = 0.0) and SA-R ($C_{rot}$ = 1.0, 2.0) are in good agreement (within 1.3–1.5% errors). In addition, the side force for SA-R ($C_{rot}$ = 1.0, 2.0) is about 3% higher than that for SA ($C_{rot}$ = 0.0).

On the other hand, in the SST model, the error between the original SST and the experiment is about 44%, while the error in the modified SST-2003 is about 0.7%, showing a significant improvement.

### 3.3 Side Force Characteristics in SA(-R) model analysis

In this section, we discuss the reason why the side force increases by about 3% for $C_{rot}$ = 1.0 or 2.0 (SA-R) compared to the case of $C_{rot}$ = 0.0 (SA). Figure 8 shows the local side force distributions in the axial direction of the vehicle, obtained from numerical calculations using the SA ($C_{rot}$ = 0.0) and SA-R ($C_{rot}$ = 1.0, 2.0) models. There is almost no difference in the local side force around the protuberance, and it can be seen that the side force for SA-R ($C_{rot}$ = 1.0, 2.0) slightly exceeds that for SA ($C_{rot}$ = 0.0) in the region where $x$/L > 0.4. (Note that the local side force distributions for $C_{rot}$ = 1.0 and 2.0 almost overlap.)

To investigate the reason for the difference in the distributions of local side force between the cases of $C_{rot}$ = 0.0 and $C_{rot} \neq 0.0$, we show in Fig. 9 the surface $C_p$ distributions on the protuberance side at the $x$/L = 0.4 cross section. It shows that there is a difference in the area where the pressure rises downstream of the flow separation on the leeward side of the vehicle. Figure 10 shows the velocity vector and $C_p$ contour at the same cross section ($x$/L = 0.4, the distribution for $C_{rot}$ = 2.0 is omitted because it is similar to that for $C_{rot}$ = 1.0). This figure shows that the separation occurs slightly more downstream for $C_{rot}$ = 0.0 than $C_{rot}$ = 1.0 or 2.0. In the case of $C_{rot}$ = 0.0, strong vorticity due to the flow around the object wall and vortices due to turbulence are not distinguished [8, 9], and thus stronger turbulent viscosity is generated than in the case of $C_{rot} \neq 0.0$. That is to say, the flow near the wall is less likely to decelerate because the energy transport between the freestream and the wall is more active, which is considered to be less likely to cause separation than $C_{rot} \neq 0.0$. On the other hand, for $C_{rot} \neq 0.0$, the turbulent viscosity of the flow around the curved geometry is suppressed by the correction in Eq. (4), resulting in pressure rise due to flow separation occurring more upstream. This causes the protuberance side to possess higher pressure than in the case of $C_{rot}$ = 0.0, which contributes to the increase in the side force.

Figure 11 shows the distribution of the surface pressure coefficient $C_p$ of the vehicle on the protuberance side at the $x$/L = 0.6 cross section, and Fig. 12 shows the visualization of the velocity vector (coloured by $C_p$) at the same cross section. According to Fig. 11, in addition to the aforementioned difference in pressure due to the "difference in the separation position of the flow around the object" (shown as (a) in the figure), the surface pressure at $C_{rot} \neq 0.0$ exceeds the surface pressure at $C_{rot}$ = 0.0 at 0.05 < $z$/(D/2) < 0.84 (region shown as (b) in the figure) on the leeward side (z > 0) of the vehicle. In addition, Fig. 12 shows that the flow from the smooth side to the protuberance side attaches for $C_{rot}$ = 0.0, while separation occurs for $C_{rot} \neq 0.0$, generating high-pressure region. (Also, as in the case of region (a) in Fig. 11, for $C_{rot} \neq 0.0$, the turbulent viscosity is suppressed, which is thought to accelerates the separation.) The formation of this high-pressure region also contributes to the increase in the side force for $C_{rot} \neq 0.0$.

### 3.4 Side Force Characteristics in SST(-2003) Model Analysis

Next, we compare the SA-R model with the original SST and SST-2003 models. For the SA-R model, the calculation result for $C_{rot}$ = 1.0 is adopted as a representative result. Figure 6 shows that the error of the side force in the original SST model is about 44% from the experimental data. On the other hand, the calculated result of SST-2003 shows good

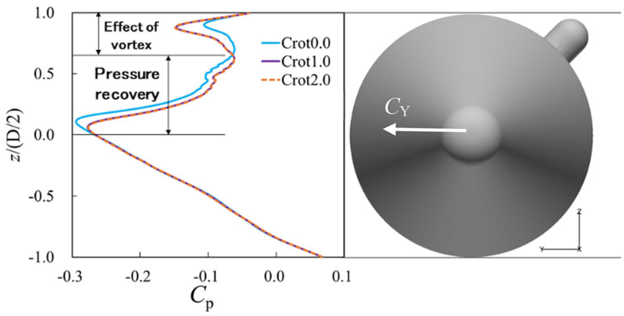



**Fig. 9** $C_p$ distribution on the protuberance side at the $x/L = 0.4$ cross section

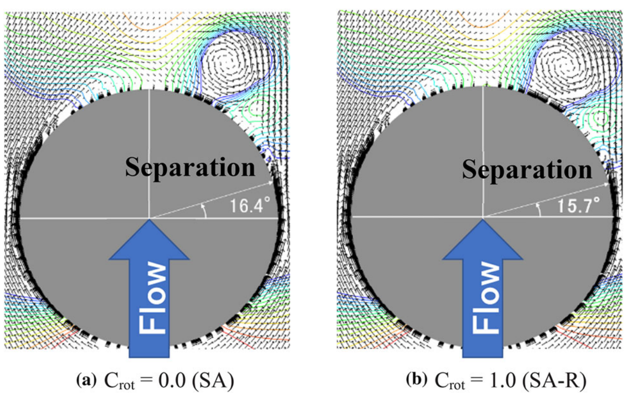



**Fig. 10** Comparison of separation points at the $x/L = 0.4$ cross section

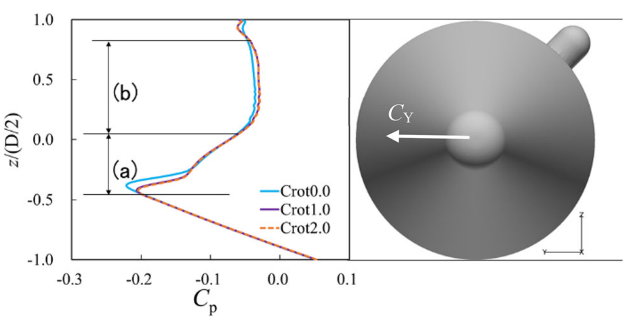



**Fig. 11** $C_p$ distribution on the protuberance side at $x/L = 0.6$ cross section

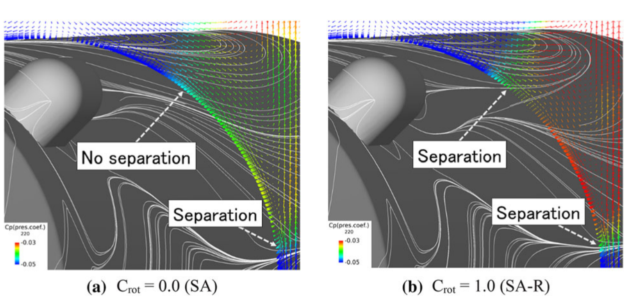



**Fig. 12** Velocity vector at $x/L = 0.6$ cross section

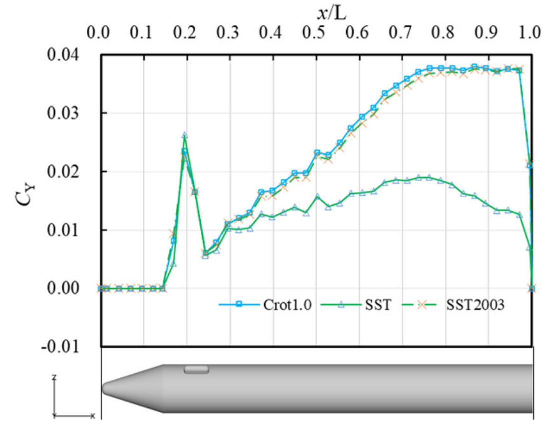



**Fig. 13** Local $C_Y$ distribution for SA-R ($C_{rot} = 1.0$) and SST (-2003) models

agreement with the experimental result with an error of about 0.7%.

Figure 13 shows the distributions of local side forces obtained from the numerical results of SA-R ($C_{rot} = 1.0$), SST and SST-2003. According to the results, the local side force of the original SST behind the protuberance ($x/L > 0.3$) is much lower than those of the other models, and the local side force begins to decrease at the aft portion of the vehicle ($x/L = 0.75$). Figure 14 shows the surface pressure distribution at the $x/L = 0.6$ cross section, and Fig. 15 shows the vorticity contours and velocity vector distribution at the same cross section. According to the Fig. 15, in the original SST, the separation vortices are dissipated on both the protuberance and smooth sides due to excessive turbulent viscosity. In particular, the vortices on the smooth side are close to the vehicle surface, and surface pressure is lowered for the SA-R and SST-2003. However, in the original SST model, the surface pressure reduction effect is lost due to the dissipation of the vortices. (This can also be seen in Fig. 14.) Also, in Fig. 15, the separation point on the protuberance side of the original SST model is much further downstream (32.6° downstream from the $z = 0$ cross section) than in the other two cases due to its excessive turbulent viscosity. It is found that the pressure on the protuberance side of the original SST is lower than that of other turbulence models because the flow separation is suppressed, and the pressure rise occurs further downstream than in other two models (see Fig. 14). From the above mechanism, the original SST underestimates the side force: The pressure difference between both sides of the vehicle is smaller due to the higher-pressure on the smooth side and lower-pressure on the protuberance side (almost cancelling each other), compared to the cases of SA-R and SST-2003 models.

On the other hand, SST-2003 model captures the pressure difference between both sides of the vehicle to the same extent as the SA-R model. This large difference between the results of the original SST and SST-2003 model can be

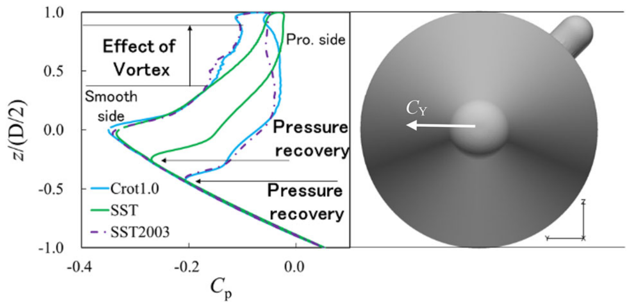



**Fig. 14** $C_p$ distribution at $x$/L = 0.6 cross section

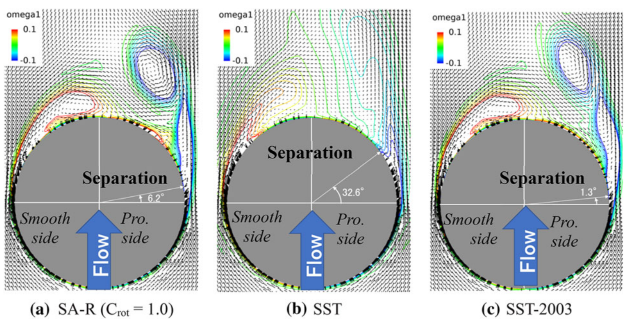



**Fig. 15** Velocity vector and vorticity at $x$/L = 0.6 cross section

attributed to the difference in the method of evaluating the turbulent viscosity coefficient $\mu_t$. Specifically, the original SST uses the vorticity $\Omega$ to evaluate $\mu_t$, and the SST-2003 uses the strain rate $S$ (see Eqs. (8) and (9) in Sec. 2.3.2). This suggests that the original SST model does not properly capture the separation because the term including $\Omega F_2$ may be inappropriately applied due to the high vorticity of the flow near the wall, even in the region where $k/\omega$ should normally be applied as turbulent viscosity in Eq. (8). On the other hand, the use of the strain rate S in the SST-2003 is considered to be an appropriate switch in the evaluation of turbulent viscosity in Eq. (9) and to properly capture the separation.

## 4 Conclusions

In this work, numerical calculations using different turbulence models were performed on the aerodynamics of a slender body with one protuberance attached to the leeward side of the front portion.

A) The effect of the parameter ($C_{rot}$) in the SA-noft2-R turbulence model

- The side force increases by about 3% for $C_{rot}$ = 1.0 and 2.0 compared to $C_{rot}$ = 0.0 (SA). There is no significant difference in the side force between the cases of $C_{rot}$ = 1.0 and 2.0.
- Calculated results of SA-R ($C_{rot}$ = 1.0 and 2.0) show more suppressed turbulence viscosity than original SA ($C_{rot}$ = 0.0), which results in less active energy transport between the uniform flow and the wall. As a result, flow separation is accelerated more in the SA-R than in the SA. Therefore, the pressure rises downstream the separation point on the protuberance side occurred more upstream, and the pressure in that region is higher than in original SA. For the same reason, on the leeward side in the SA-R case, the flow from the smooth side to the protuberance separates, and a high-pressure region is formed on the protuberance side.
- As described above, the calculated side force in the SA-R (both cases of $C_{rot}$ = 1.0 and 2.0) slightly increases due to the high-pressure on the protuberance side because of its ease of flow separation.

B) Comparison of SST models (original SST, SST-2003)

- The calculated side force in the original SST is much smaller than in other turbulence models and the error from experiment is much larger (about 44%). On the other hand, the calculation result of SST-2003 shows an error of about 0.7% from the experiment, and the accuracy of the calculation result for the side force is about the same as the SA-R model. In the calculation result of the original SST model,
- Vortices on the leeward of the vehicle dissipates due to excessive turbulent viscosity.
- Due to the dissipation of the vortices, the low-pressure region caused by the proximity of the vortices disappears, and the smooth side of the vehicle possesses higher pressure than the results of other models.
- Excessive turbulent viscosity on the protuberance side causes flow separation farther downstream than in other models. Therefore, the accelerated expansion flow near the wall does not decelerate, and the surface pressure on the protuberance side becomes low.
- From the above mechanisms, the pressure difference between both sides of the vehicle is underestimated in the original SST, and the side force is much smaller than in other models.

The SST-2003 model uses strain rate instead of vorticity to evaluate turbulent viscosity, which prevents the generation of excessive turbulent viscosity and suppresses the dissipation of vortices. In other words, the SST-2003 model can properly capture the flow separation caused by the curvature of the object shape, even behind the shock wave.

## 5 Appendices

### 5.1 A. Axial Force Coefficient, $C_A$

Figure 16 shows the calculated and experimental values of the axial force coefficient $C_A$. The calculated results are

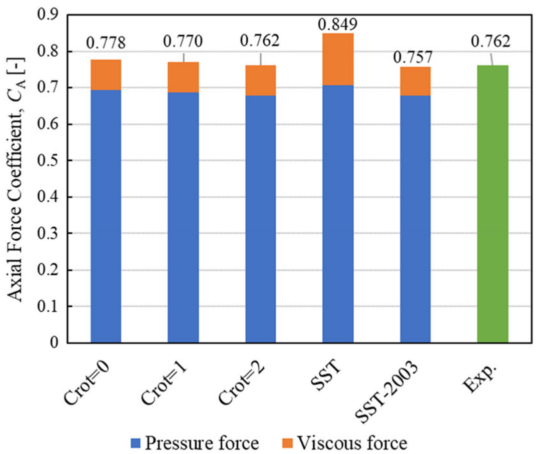



**Fig. 16** Axial force coefficients $C_A$

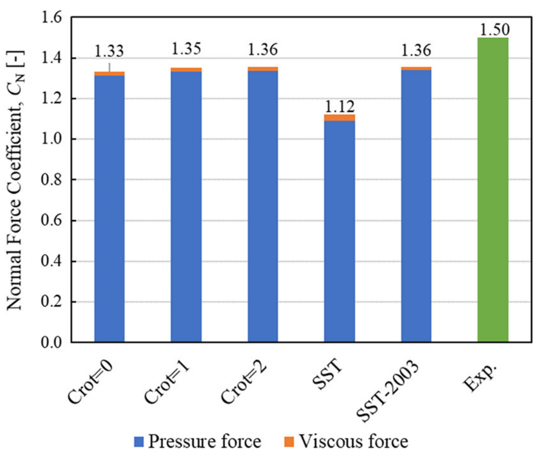



**Fig. 17** Normal force coefficients $C_N$

color-coded for the pressure and viscous components. According to the figure, the calculated axial forces by each turbulent model other than the original SST have small errors compared with the experiment [SA ($C_{rot} = 0.0$): 2.14%, SA-R ($C_{rot} = 1.0$): 1.02%, SA-R ($C_{rot} = 2.0$): 0.02%, SST-2003: 0.684%], especially for the SA-R ($C_{rot} = 2.0$) and SST-2003 demonstrating superior performance in axial force prediction. On the other hand, original SST has a larger error from experiment (SST: 11.4%) than the other turbulent models, which may be due to an overestimation of viscous force.

### 5.2 Normal Force Coefficient, $C_N$

Figure 17 shows the calculated and experimental results of the normal force coefficient $C_N$. Overall, each turbulence model tends to underestimate the normal force of the experiment, and it is difficult to decide which model is the best whereas the SST obviously failed in the normal force prediction. The original SST underestimates the normal force even more than the other turbulent models, possibly because the force that pulls the vehicle toward the vortex core (like the vortex lift force in a delta wing) is reduced due to vortex dissipation caused by excessive turbulent viscosity, as shown in Fig. 6. Further discussions on the detailed mechanism are needed and left as the future work.

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